EventAI: A Multi-Agent, Retrieval-Augmented Pipeline for Turning Event Recordings into Publish-Ready Content, with a Live Evaluation on TED Talks

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***Abstract*—Event recordings are rarely turned into the summaries, articles and social-media posts that organisers need, because doing so by hand is slow and generic chatbots require manual transcription, prompting and fact-checking. We present EventAI, a web application that converts a recording into six publish-ready formats through Gemini speech-to-text on Vertex AI, a LangGraph multi-agent pipeline with six parallel format-specialised writers, a batched grounding and moderation guardrail with targeted regeneration, selective retrieval-augmented generation (RAG) for long talks, Hindi translation and mandatory human approval. We evaluate the unmodified system live on 18 full TED talks (4.0 h) with human reference transcripts. EventAI transcribes at 2.19% corpus word error rate and produces all drafts plus translations in a median of 90.2 s per talk (real-time factor 0.12). Its embedding retriever ranks the relevant passage first for 76% of 109 mechanically labelled questions (MRR 0.87, versus 0.73 for BM25 and 0.41 for the previous hashed vectors). In a controlled ablation on eleven long talks, RAG raises coverage of key points from the unseen second half of a talk from 46.8% to 68.4% ($p = 0.0020$), while overall coverage is unchanged, i.e. it rebalances rather than enlarges the content. Two LLM judges rate the drafts 4.96 and 4.91 out of 5 for factual consistency. Live runs and fault injection also exposed concrete defects—an inert circuit breaker, sticky model fallback, silent offline fallback, few-shot template leakage and an unguarded response shape—which we report together with their fixes and the re-measured results.**

***Index Terms*—multi-agent systems, retrieval-augmented generation, speech recognition, large language models, content generation, LLM-as-a-judge, human-in-the-loop**

I. Introduction

Conferences, meetups and college events are now routinely recorded, yet most recordings are never turned into the written material that organisers need afterwards: a summary for stakeholders, a blog article, posts for LinkedIn and Instagram, a newsletter and a flyer for the next edition. Producing these by hand takes hours per session, and generic large language model (LLM) chat interfaces do not solve the problem end to end: the user must transcribe the talk, paste text in pieces, prompt each format separately and check every claim against the recording [1].

This paper presents *EventAI*, an open web application that converts an event recording into six publish-ready content formats with a human approval step. EventAI combines Gemini speech-to-text on Google Cloud Vertex AI, a LangGraph [2] multi-agent pipeline (analysis, planning, six format-specialised writer agents running in parallel, a guardrail and translation), *selective* retrieval-augmented generation (RAG) [3] that supplies the parts of long talks the writers cannot see directly, and a review workflow in which no draft is marked ready to publish without approval.

The contributions of this implementation paper are:

1) The design and implementation of an end-to-end, human-in-the-loop recording-to-content system with a stateful multi-agent graph, a batched LLM guardrail with targeted regeneration of only the failing drafts, and a selective RAG stage driven by the editorial brief (Section III).
2) A reproducible, cost-aware evaluation on 18 full TED talks (about four hours of audio) with human reference transcripts [4], covering speech recognition accuracy, stage latency, token usage, retrieval quality, a controlled RAG ablation, generated-content quality and reliability under injected faults (Sections IV–V).
3) An honest account of the defects that the evaluation itself exposed in the system (guardrail scope, few-shot template leakage, an inert circuit breaker, sticky model fallback, silent offline fallback and an unguarded JSON shape), how each was found, fixed and re-measured (Section VI).

All raw measurements, scripts and figures are released with the code so that every number in this paper can be regenerated.

II. Related Work

Speech recognition for long-form talks. Large weakly supervised models such as Whisper [5] made robust transcription of lectures and talks practical; the ESB benchmark [6] standardised text normalisation for comparing systems across datasets, and TED-LIUM 3 [4] remains a standard corpus of real conference talks with human transcripts. Multimodal LLMs such as Gemini [7] accept audio directly, which lets a single model family serve both transcription and generation.

Summarising spoken content. Meeting and lecture summarisation has been studied as query-focused and long-input summarisation [8]. Long inputs remain difficult: models underuse information in the middle and end of long contexts [9], which motivates retrieving relevant passages instead of relying on truncated context.

Retrieval-augmented generation. RAG conditions generation on retrieved passages [3]. Dense retrievers [10] and modern general embedding models [11] usually outperform lexical BM25 [12], although BM25 remains a strong baseline. Retrieval quality is measured with Recall@k, MRR and nDCG [13]; frameworks such as RAGAS [14] evaluate the generation side.

Multi-agent LLM systems and guardrails. Tool-using and multi-agent patterns such as ReAct [15] and AutoGen [16] decompose complex tasks into cooperating LLM roles; LangGraph [2] expresses such workflows as explicit stateful graphs with conditional routing. Because LLMs hallucinate [1] and can produce harmful text [17], production systems add verification steps; LLM-as-a-judge [18], [19] is a common automatic proxy for human evaluation, with known biases that we mitigate by using a different and larger judge model and by measuring inter-judge agreement. Retry with back-off and the circuit-breaker pattern [20] are standard resilience techniques for calling remote services.

EventAI combines these strands in one deployed application and evaluates them together on the same recordings, reporting accuracy, latency, token usage and failure behaviour of the whole recording-to-publication pipeline.

III. System Design and Implementation

Fig. 1 shows the architecture. A React 19/TypeScript client talks to an asynchronous FastAPI backend over REST and Server-Sent Events (SSE); uploads are streamed to local disk or S3-compatible object storage (Cloudflare R2, S3, MinIO) with SHA-256 integrity hashes; relational data (users, events, jobs, transcripts, content, embeddings, audit log) is kept in PostgreSQL or SQLite. Jobs run as in-process background tasks or Celery workers. All model calls go to Google Cloud Vertex AI using Application Default Credentials, so no API key is stored in the application.

A. Ingestion and speech-to-text

An organiser uploads audio, video or a text transcript after confirming that participants consented to processing. The job runner validates the file (extension whitelist, size and duration limits, optional ClamAV scan), measures duration, sample rate and silence ratio, and applies a voice-activity pre-check that rejects recordings with no detectable speech. Transcription uses Gemini on Vertex AI. Because Vertex accepts at most about 20 MB of inline audio per request, the recording is resampled to 16 kHz mono and split into windows of 300 s; each window end is moved to the quietest 250 ms frame within the preceding 10 s so that boundaries fall between words. Chunks are transcribed concurrently (four at a time) and concatenated. If the engine fails, the runner tries every other configured engine (Gemini API, VideoDB) in turn, skipping engines whose credentials were rejected at start-up.

B. Multi-agent pipeline

The pipeline is a LangGraph `StateGraph` (Fig. 2). A quality gate routes low-confidence transcripts to `needs_reupload`. An *analysis* node extracts topics, highlights, speaker characteristics, tone and an editorial brief (angle, key messages, audience, verbatim quotes) in one structured JSON call; a *planner* node reuses this brief. After selective RAG (Section III-D), the LangGraph `Send` API fans out to the six requested writer agents in parallel: executive summary, blog article, LinkedIn post, newsletter, Instagram caption and event flyer. Each agent is defined by a Markdown *skill* file (persona, structure, do/don't rules, an example) plus the author's profile (name, organisation, tone, sign-off). Writers see the first 12,000 characters of the transcript, the brief and the retrieved excerpts, and must return JSON.

C. Guardrail and human review

A single batched LLM call checks all drafts at once for *grounding* (claims not supported by the transcript) and *moderation*. The guardrail receives the full transcript and the author profile, so that user-supplied sign-offs are not flagged as invented. Drafts that fail are regenerated once—only the failing types, skipping the planner—and drafts that still fail are routed to human review; otherwise the job proceeds to translation, in which each target language is one batched call for all drafts. Every draft then waits for approval in the Content Studio, where it can be edited, approved, copied or exported; nothing is marked ready to publish automatically.

D. Selective retrieval-augmented generation

Transcripts are split into overlapping 250-word chunks (50-word overlap) and embedded with `gemini-embedding-001` (768 dimensions); vectors are stored in a JSON column, so no database extension is needed. Retrieval is *selective*: it runs only when the transcript exceeds what the writers see directly (>2,000 words) or when the organiser enables organisation memory. The candidate pool excludes chunks already inside the writers' 12,000-character excerpt and, with organisation memory, adds chunks

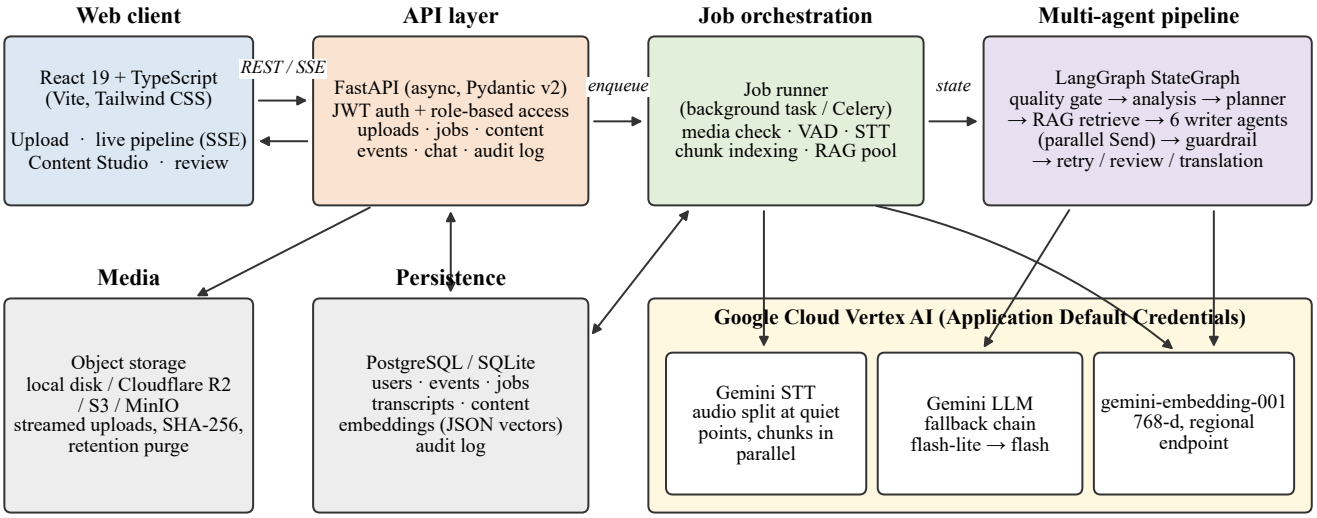


Fig. 1. EventAI architecture. Arrows show request and data flow; all model calls use Vertex AI with Application Default Credentials.

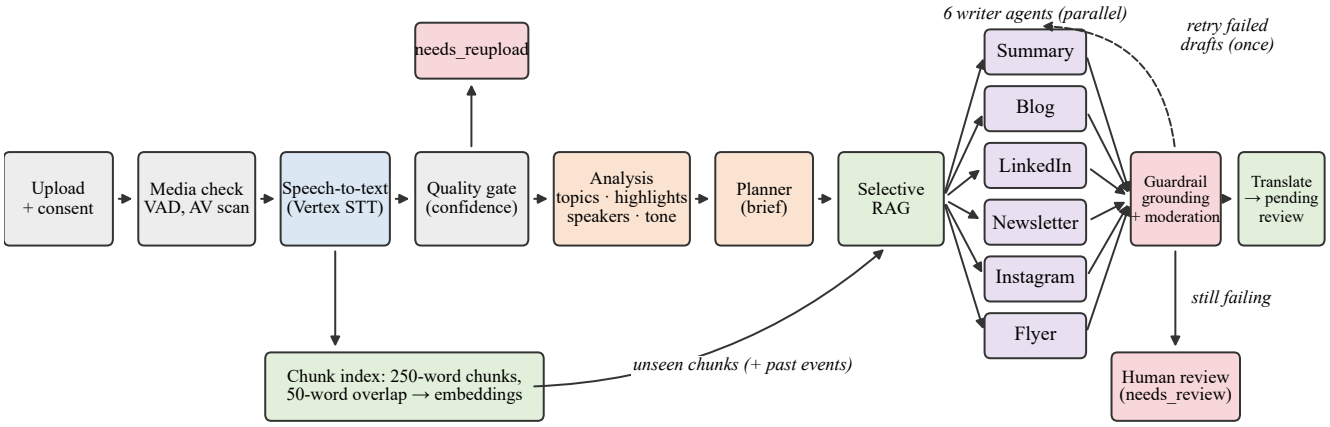


Fig. 2. LangGraph pipeline. Only drafts that fail the guardrail are regenerated, once; drafts still failing are routed to human review, the rest are translated.

from the organiser's earlier events. A `rag_retrieve` graph node builds the query from the editorial brief (angle, key messages, topics and highlights) and passes the top $k{=}6$ chunks, by cosine similarity, to every writer, labelled as later parts of this session or as earlier events that must not be presented as said in this session. If the embedding service is unavailable, a hashed bag-of-words representation is used instead.

E. Resilience

Every model call has a timeout (60 s) and an immediate retry for transient errors. The Vertex provider always starts with the configured model and walks a fallback chain (`gemini-3.1-flash-lite`, then `gemini-3.6-flash`) when a model returns an error, a permanent "not found" or an empty response. Each model has its own circuit breaker: after three consecutive failed calls the model is skipped for 60 s and then tried again; a model that answered "not found" is skipped for the rest of the process. Every model reply is coerced to a JSON object. If every model fails, an offline template generator keeps the job running, but the job is flagged as degraded and routed to human review. Provider credentials are checked at start-up, and speech-to-text falls back through every configured engine whose credentials were accepted. This is the version after the fixes of Section VI; the differences to the version used in most live runs are listed there.

F. Implementation

The backend is Python 3.11 (FastAPI 0.115, LangGraph 0.3.31, SQLAlchemy 2.0, google-genai); the frontend is React 19, TypeScript and Vite with Tailwind CSS. The system is configured by environment variables (provider, models, RAG threshold, k, embedding region) and ships with Docker Compose files.

IV. Evaluation Methodology

We follow a hybrid, cost-aware protocol: automated local tests with mocked providers, a pilot on two talks, live end-to-end runs on the whole dataset, cached-stage experiments that reuse stored transcripts, and mock-based fault injection. Every result is labelled as live, cached-stage, mock, dataset-based or automatic-judge.

A. Dataset

We use the long-form release of TED-LIUM 3 [4] (CC BY-NC-ND 3.0, used for non-commercial academic evaluation only): 19 complete TED talks whose official segments are concatenated per talk, each with its human reference transcript. One 5.5 s item is excluded as it is not a talk, leaving 18 talks (4.0 h; 3.1–25.1 min, median 15.2 min) of which 11 exceed 2,000 words and therefore trigger selective RAG. TED talks are recorded conference presentations, i.e. the target domain of EventAI.

B. Execution environment

All live runs use the unmodified backend, driven in-process through its HTTP routes (`POST /uploads`, `POST /jobs/{id}/regenerate`) by a harness that only wraps functions to record wall-clock timings and the token counts returned by the SDK. Each worker uses its own SQLite database. Configuration: Vertex AI (`global` endpoint for generation, `asia-south1` for embeddings), `gemini-3.1-flash-lite` for transcription, analysis, writing, guardrail and translation, `gemini-embedding-001` for RAG, six formats in English plus Hindi translation, and a fixed neutral author profile. The provider's in-process state (model memory, circuit breakers) is reset before each run so that every run starts on the configured model. The live runs used the version before the fixes of Section VI, except the seven runs of *Elizabeth Gilbert*, which failed with that version and were repeated after the response-shape fix. The fixes change behaviour only when calls fail or replies are malformed; in the other runs the only such events were the failed calls during network outages (Section V-G). Two jobs ran concurrently on a Windows 11 laptop over a home broadband connection. Runs affected by machine-wide network outages (a single call stalled for >600 s, or connections refused) are excluded from latency statistics, reported in Section V, and re-run where needed.

C. Experiments and metrics

E1 Software tests. The project's pytest suite (offline fake providers) with line coverage.

E2 Speech recognition. Word and character error rate (WER/CER) against the human references after the Whisper English normaliser [5], as in the ESB protocol [6], with `<unk>` tokens removed from references; we report corpus WER ($\sum(S+D+I)/\sum N$), per-talk WER and the real-time factor (RTF, processing time / audio duration). A cached-stage experiment compares three Gemini models and three chunk lengths using the app's own `transcribe_file` function.

E3 Latency and tokens. Per-stage wall time and token usage of the 18 live end-to-end runs (upload to finished drafts).

E4 Retrieval. Human reference transcripts of the 11 long talks are chunked exactly as in the app. For each talk the judge model writes ten factual questions, each with a verbatim supporting quote; a question is kept only if the quote occurs verbatim in the reference, and the relevant chunks are those containing it, so relevance is determined mechanically. We compare `gemini-embedding-001`, the hashed bag-of-words vectors used by the previous version of EventAI, and Okapi BM25 [12] with Recall@k, MRR and nDCG@6 [13], both within a talk (the default) and across all talks (organisation-memory setting).

E5 RAG ablation. For each long talk, the stored transcript is regenerated three times with RAG on and three times with RAG off, alternating conditions. The judge lists ten key points per talk from the human reference with verbatim quotes; a point is *late* if its quote lies beyond the share of the talk the writers see directly. Coverage of early and late points per draft is averaged per talk and condition and compared with the paired Wilcoxon signed-rank test [21] over talks.

E6 Content quality. Each English draft of the end-to-end runs is scored 1–5 on factual consistency, relevance, completeness, readability and instruction adherence, with a list of unsupported claims, by `gemini-3.7-flash` at temperature 0 against the human reference transcript (G-Eval-style LLM-as-a-judge [18], [19]), blind to the condition. A second judge (`gemini-3.6-flash`) re-scores the end-to-end runs; we report exact agreement, agreement within one point and quadratic weighted κ [22]. Deterministic format checks (word limits, hashtag counts, banned phrases, invented dates) complement the judge.

E7 Reliability. *Live*: API errors, empty responses, model fallbacks, guardrail retries and job outcomes over all live runs. *Mock*: nine fault scenarios $\times$ 20 trials injected into the production Vertex provider and retry code through a fake client, keeping the real back-off sleeps; plus probes of the circuit breaker and of a sustained rate limit on the primary model. The mock experiment was run before and after the fixes.

E8 Usability walkthrough. The web client is driven with Playwright (Microsoft Edge) through registration, upload, the live pipeline view, the Content Studio and review, on a TED talk, to document the user-facing workflow (Fig. 12).

TABLE I
AUTOMATED TEST SUITE (PYTEST, OFFLINE FAKE PROVIDERS).

Item	Value
Test cases (files)	80 (14)
Passed / failed / errors	80 / 0 / 0
Line coverage (app)	65.9% of 3644 statements
Coverage: core / graphs / services	93.8 / 66.8 / 54.1%
Coverage: providers / api	62.7 / 59.5%
Wall time	41.9 s

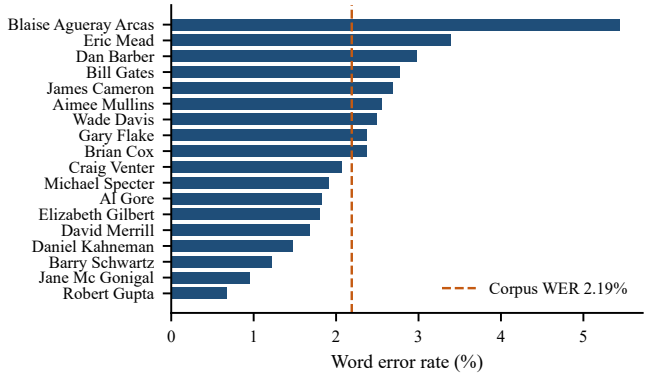


Fig. 3. Per-talk WER of the production configuration (live pipeline, 18 talks).

V. RESULTS

All numbers below are generated by the released analysis script from saved raw results.

A. E1: Software tests

The suite (Table I) passes completely; it includes ten regression tests added for the fixes described in Section VI (circuit breaker, model selection, response shape, degraded-output flag, credential check and speech-to-text fallback chain). Coverage is lowest for the speech-to-text providers and the job runner's media path, which the offline suite deliberately does not call; those paths are exercised by the live experiments below.

B. E2: Speech recognition

In the live pipeline, `gemini-3.1-flash-lite` with 300 s quiet-point chunks reaches a corpus WER of 2.19% over 44,283 reference words (Table II); per-talk WER ranges from 0.68% to 5.45% (Fig. 3) and the median real-time factor is 0.048; the longest (25-minute) talk was transcribed in 54 s. Chunk length barely affects accuracy (2.19%, 2.19% and 2.23% for 120, 300 and 600 s windows), which indicates that the quiet-point splitting does not cut words. The larger `gemini-3.6-flash` lowers corpus WER only slightly, to 2.10%. The newest `gemini-3.7-flash` is the most accurate (1.98%) but more than twice as slow (median RTF 0.106, 90.7 s per talk), and 10 of its 28 transcription attempts failed—in three talks chunks exceeded the 240 s per-chunk timeout, the others coincided with network outages—and were repeated until all 18 talks succeeded. The production choice of `flash-lite` therefore trades 0.2 WER points for less than half the transcription time (Fig. 4). Across talks, WER is not related to the judged quality of the drafts (Spearman $\rho = -0.09$, $n = 18$), consistent with the low error rate. Substitutions are the most frequent error type (464, against 224 deletions and 284 insertions). Transcription consumed 1,500–2,000 input tokens per minute of audio, depending on the chunk length.

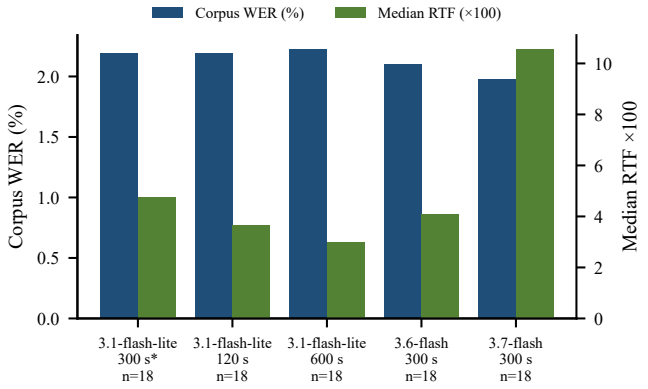


Fig. 4. Corpus WER and median real-time factor by model and chunk length (* = live pipeline).

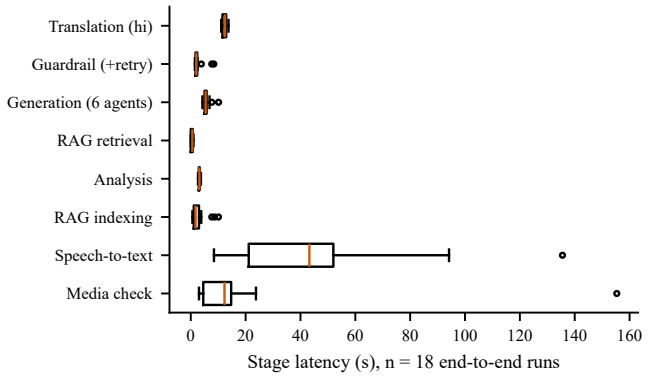


Fig. 5. Stage latency distribution of the live end-to-end runs.

C. E3: Latency and token usage

A completed job—transcription, analysis, six drafts, guardrail and a Hindi translation of all drafts—took a median of 90.2 s (IQR 56.4–98.2 s) for talks with a median length of 15.2 min, i.e. a median end-to-end real-time factor of 0.12 (Table III, Fig. 5). Speech-to-text dominates (median 43.2 s), followed by the local media check (local silence analysis, median 12.3 s, with one outlier of 155 s) and translation (12.5 s). Thanks to the parallel fan-out, writing all six drafts takes only 5.4 s (median); analysis (3.1 s), the guardrail (2.0 s) and RAG retrieval (0.6 s) are minor. Total latency grows with talk length (Spearman $\rho = 0.77$, $p = 1.8 \times 10^{-4}$, $n = 18$; Fig. 6). A job made on average 12.8 LLM calls and used 64.5 k input and 12.0 k output tokens, of which the six writers

TABLE II

SPEECH RECOGNITION ON TED-LIUM 3 TALKS (WER/CER IN %, WHISPER NORMALISER). THE FIRST ROW IS THE PRODUCTION CONFIGURATION INSIDE THE LIVE PIPELINE; THE OTHERS ARE CACHED-STAGE RUNS OF THE SAME `TRANSCRIBE_FILE` FUNCTION, FOUR CONFIGURATIONS PROCESSED CONCURRENTLY.

Model	Chunk (s)	Talks	Corpus WER	Median	Max	CER	S/D/I	RTF	STT (s)
3.1-flash-lite (pipeline)	300	18	2.19	2.22	5.45	1.15	464/224/284	0.048	43.2
3.1-flash-lite	120	18	2.19	2.17	5.03	1.17	460/229/279	0.037	25.5
3.1-flash-lite	600	18	2.23	2.34	5.38	1.17	463/227/297	0.030	24.1
3.6-flash	300	18	2.10	1.89	5.10	1.09	413/242/276	0.041	30.8
3.7-flash	300	18	1.98	1.87	4.54	1.06	380/210/285	0.106	90.7

TABLE III

PER-STAGE LATENCY OF THE LIVE END-TO-END RUNS (SECONDS; COMPLETED JOBS, UPLOAD TO FINISHED DRAFTS INCL. HINDI TRANSLATION).

Stage	n	Median	IQR	Range
Media check	18	12.30	4.55–14.72	3.00–155.35
Speech-to-text	18	43.24	21.13–51.99	8.45–135.53
RAG indexing	18	1.74	1.09–2.99	0.60–10.09
Analysis	18	3.14	2.92–3.27	2.64–3.70
RAG retrieval	18	0.57	0.00–0.73	0.00–0.99
Generation (6 agents)	18	5.41	5.02–5.78	4.25–10.13
Guardrail (+retry)	18	2.04	1.73–2.37	1.49–8.40
Translation (hi)	18	12.49	11.55–12.92	11.10–13.75
Total (upload to drafts)	18	90.20	56.40–98.20	32.90–226.30

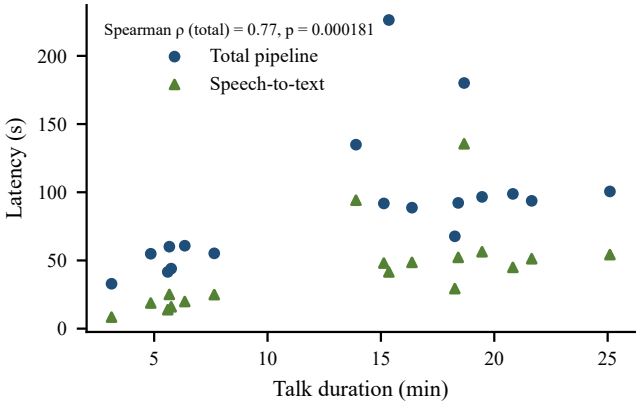


Fig. 6. Talk duration versus total and speech-to-text latency.

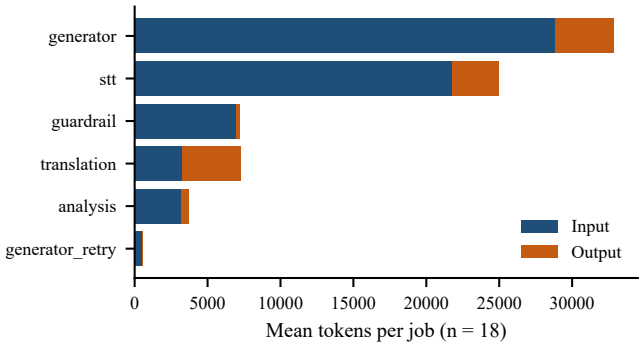


Fig. 7. Mean tokens per job by stage (live end-to-end runs).

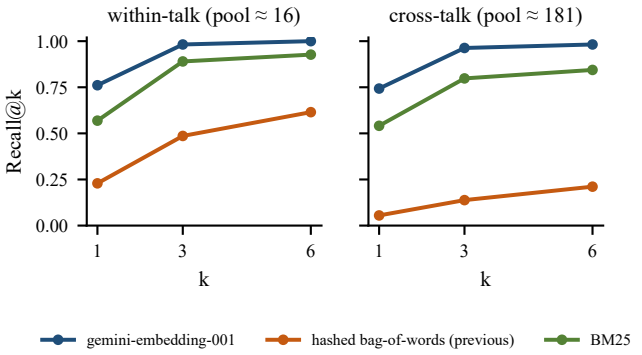


Fig. 8. Recall@k of the three retrievers within a talk and across all talks.

account for 28.8 k input tokens and transcription for 21.8 k audio tokens (Fig. 7).

D. E4: Retrieval quality

Of 110 generated questions, 109 had a quote found verbatim in the reference and were kept. With `gemini-embedding-001`, the relevant chunk is ranked first for 76% of the questions and is always within the top six that EventAI passes to the writers (Recall@6 = 1.00, MRR 0.87; Table IV, Fig. 8). BM25 is a strong lexical baseline (MRR 0.73), while the hashed bag-of-words vectors of the previous EventAI version reach only MRR 0.41. The gap widens in the organisation-memory setting with 181 candidate chunks from eleven talks: the embedding retriever keeps Recall@6 at 0.98, whereas hashed vectors collapse to 0.21, i.e. they could not have supported cross-event memory.

E. E5: Effect of selective RAG

On the eleven long talks, RAG raised coverage of key points that lie beyond the writers' 12,000-character excerpt from 46.8% to 68.4% (+21.6 points; 10 of 11 talks improved, none worse; $p = 0.0020$), and from 50.8% to 74.7% for the long formats (Table V, Fig. 9). Coverage of early points fell from 70.1% to 58.5% ($p = 0.0029$), and total coverage was unchanged (59.8% vs. 59.1%, $p = 0.71$): with a fixed output length, RAG *redistributes* the drafts' attention over the whole talk rather than adding content. Rubric scores, including factual consistency, did not change significantly, and unsupported claims stayed rare in both conditions (0.01 vs.

TABLE IV
RETRIEVAL ON 109 LABELLED QUESTIONS OVER 181 CHUNKS OF 11 LONG TALKS.

Pool	Retriever	R@1	R@3	R@6	MRR	nDCG@6
within-talk	gemini-embedding-001	0.76	0.98	1.00	0.87	0.90
within-talk	hashed bag-of-words (prev.)	0.23	0.49	0.61	0.41	0.41
within-talk	BM25	0.57	0.89	0.93	0.73	0.77
cross-talk	gemini-embedding-001	0.74	0.96	0.98	0.85	0.88
cross-talk	hashed bag-of-words (prev.)	0.06	0.14	0.21	0.13	0.12
cross-talk	BM25	0.54	0.80	0.84	0.68	0.71

TABLE V
RAG ABLATION: SAME STORED TRANSCRIPT, 3 RUNS PER CONDITION, AVERAGED PER TALK ($n = 11$ LONG TALKS); TWO-SIDED WILCOXON SIGNED-RANK TEST. COVERAGE FROM THE AUTOMATIC JUDGE.

Metric	RAG off	RAG on	Δ	+/−	p
Late coverage, all formats (%)	46.8	68.4	+21.6	10/0	0.0020
Early coverage, all formats (%)	70.1	58.5	-11.6	1/10	0.0029
Late coverage, long formats (%)	50.8	74.7	+23.9	9/0	0.0039
All key points, all formats (%)	59.8	59.1	-0.7	5/5	0.7148
Factual cons.	4.96	4.99	+0.03	4/1	0.1875
Relevance	5.00	4.97	-0.03	0/2	0.5000
Completeness	4.20	4.28	+0.08	7/3	0.3223
Readability	5.00	4.96	-0.03	1/2	0.5000
Instr. adherence	4.67	4.69	+0.02	4/6	0.7090
Unsupported claims per draft	0.04	0.01	-0.04	1/5	0.1250

+/−: talks whose per-talk mean rose/fell with RAG.

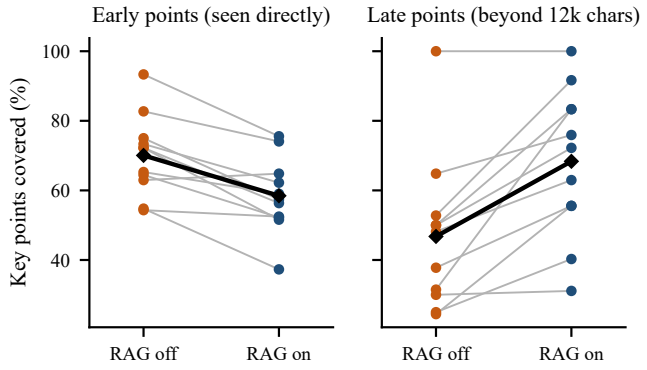


Fig. 9. Key-point coverage per talk without and with RAG (grey lines: talks; black: mean).

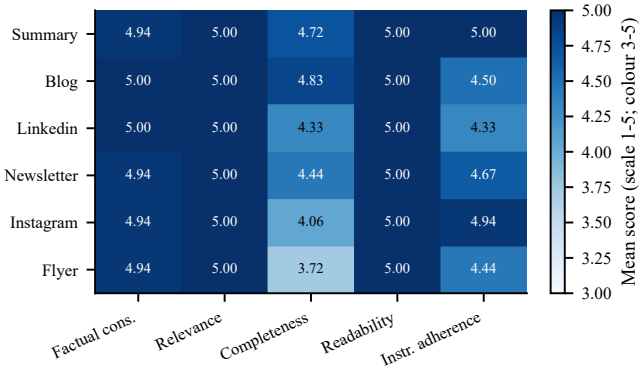


Fig. 10. Mean judge score per format and dimension (colour scale 3–5).

0.04 per draft). The *Elizabeth Gilbert* runs were made after the response-shape fix (Section VI); without them the result is the same in direction and significance (late coverage 48.4% to 72.1%, $p = 0.0039$, $n = 10$). For organisers this means that, with RAG, posts about long talks no longer ignore their second half.

F. E6: Content quality

Judged against the human transcript, the 108 English drafts of the 18 end-to-end runs score 4.96 for factual consistency with 0–0.06 unsupported claims per draft, 5.00 for relevance and readability, 4.65 for instruction adherence and 4.35 for completeness (Table VI, Fig. 10). Completeness is lowest for the short formats—Instagram (4.06) and especially the flyer (3.72)—which is expected from their length limits. The deterministic checks (Table VII) confirm summary length, Instagram hashtag counts, absence of banned LinkedIn clichés and absence of invented event dates in all drafts, but only 12 of 18 LinkedIn posts respect the 3–5 hashtag rule. The two judges agree within one point on every item and exactly on 73.1–100% of items per dimension (Table VIII); weighted κ is 0.62 for completeness and 0.88 for instruction adherence, and is low or undefined for relevance, readability and factual consistency only because almost all scores are 5 (the well-known κ paradox at a ceiling). The high scores should therefore be read as "no defects detected by two LLM judges" rather than as human ratings.

G. E7: Reliability

Live. Across 93 live pipeline runs (84 planned, two repeated after outages and the seven *Elizabeth Gilbert* runs repeated after the response-shape fix) the backend made 946 model and embedding calls; 22 failed (2.3%): 2 rate-limit (429) responses, 7 timeouts and 13 network errors during host outages. No empty responses and no model fallbacks occurred. The guardrail triggered a targeted regeneration in 8 runs; after it, 2 of 510 final English drafts (both newsletters) remained flagged and their jobs were routed to human review. 83 jobs reached `pending_review`, 2 `needs_review` and 8 failed: the 7 *Elizabeth Gilbert* runs before the response-shape fix (all 7 repeated runs after it completed) and 1 from a network outage combined with an invalid fallback STT credential.

TABLE VI
AUTOMATIC-JUDGE QUALITY SCORES (1–5) OF THE ENGLISH DRAFTS OF THE LIVE END-TO-END RUNS, JUDGED AGAINST THE HUMAN REFERENCE TRANSCRIPT.

Format	n	Fact.	Rel.	Compl.	Read.	Instr.	Unsup.
Summary	18	4.94	5.00	4.72	5.00	5.00	0.06
Blog	18	5.00	5.00	4.83	5.00	4.50	0.00
Linkedin	18	5.00	5.00	4.33	5.00	4.33	0.00
Newsletter	18	4.94	5.00	4.44	5.00	4.67	0.06
Instagram	18	4.94	5.00	4.06	5.00	4.94	0.06
Flyer	18	4.94	5.00	3.72	5.00	4.44	0.06
All	108	4.96	5.00	4.35	5.00	4.65	–

TABLE VII
DETERMINISTIC FORMAT CHECKS ON THE END-TO-END DRAFTS.

Check	Pass
linkedin 3-5 hashtags	12/18
linkedin no banned phrases	18/18
instagram 10-15 hashtags	18/18
summary <= 400 words	18/18
flyer no invented date (TBA or none)	18/18

TABLE VIII
AGREEMENT BETWEEN THE PRIMARY JUDGE (J1, GEMINI-3.7-FLASH) AND A SECOND JUDGE (J2, GEMINI-3.6-FLASH) ON THE SAME DRAFTS.

Dimension	n	Exact %	±1 %	QWK	J1	J2
Factual cons.	108	90.7	100.0	0.25	4.96	4.91
Relevance	108	100.0	100.0	–	5.00	5.00
Completeness	108	73.1	100.0	0.62	4.35	4.32
Readability	108	99.1	100.0	0.00	5.00	4.99
Instr. adherence	108	89.8	100.0	0.88	4.65	4.69

Mock. Transient 429s, timeouts, unknown models, empty responses and fenced JSON are all recovered with usable output (Table IX, Fig. 11); persistent rate limiting costs about 1 s before the fallback model answers. Prose without any JSON is still passed on unparsed. Before the fixes, the next call after an empty response or a persistent 429 stayed on the fallback model (sticky fallback), the circuit breaker short-circuited none of 8 consecutive failing calls, and offline template text was returned without any flag. After them, the next call returns to the primary model, the breaker rejects 3 of the 8 calls, a persistently rate-limited primary is skipped from the fourth call on, so later calls no longer pay the 1 s, and all 20 trials in which every model failed are flagged as degraded.

H. E8: User-facing workflow

Fig. 12 shows the workflow captured by Playwright on the 3-minute *Brian Cox* talk: an organiser selects formats, languages and confirms consent, follows the pipeline stages live and reviews, edits, approves or rejects each draft. In this run the guardrail flagged the summary and the job correctly stopped at `needs_review`. The walkthrough also exposed a stale label in the client (the transcription stage still named VideoDB, which the deployment no longer uses); it was corrected after the walkthrough.

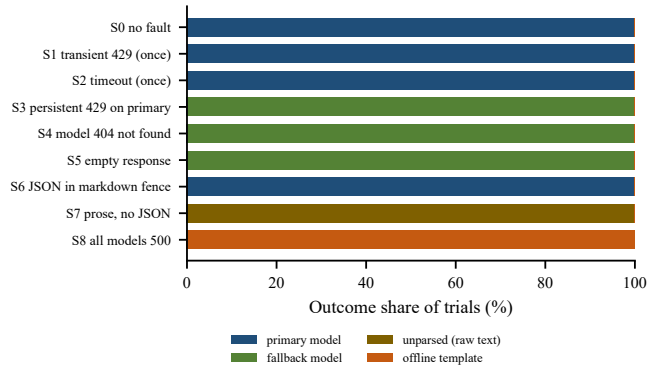


Fig. 11. Outcomes of mock fault injection (20 trials per scenario).

VI. DISCUSSION

A. Defects exposed by the evaluation

Running the real system on real talks surfaced problems that the offline test suite, which uses fake providers, could not reveal. We report them because they are the most practically useful outcome for practitioners building similar pipelines.

Guardrail scope (fixed before the final runs). The guardrail originally saw only the first 12,000 transcript characters and not the author profile. In the pilot, all six drafts of a talk were flagged: user-supplied sign-offs were reported as "invented", and correct facts from late in the talk (supplied to the writers by RAG) would have been penalised. The guardrail now receives the whole transcript and the profile.

Few-shot template leakage (fixed). The flyer and summary skill files contained example outputs with a concrete date ("October 24, 2026") and a venue, and the flyer rules demanded a date. With no event metadata available, writers copied the example date into drafts; the guardrail caught it, but only after a regeneration round-trip. A shared rule forbidding copying of example details and requiring "TBA" for unknown logistics removed the behaviour (Table VII).

Sticky model fallback (fixed after the evaluation). The Vertex provider remembered the last working model for the whole process. In the pilot, a single empty response from `gemini-3.1-flash-lite` switched every subsequent call of the process to the larger `gemini-3.6-flash`, silently changing quality, latency and cost. The provider now always

TABLE IX
MOCK-BASED FAULT INJECTION INTO THE PRODUCTION VERTEX PROVIDER (20 TRIALS PER SCENARIO). "NEXT CALL" IS THE FIRST MODEL TRIED BY THE NEXT CALL IN THE SAME PROCESS; "FLAGGED" COUNTS TRIALS WHOSE FALLBACK OUTPUT WAS MARKED AS DEGRADED.

Scenario	Outcome (of 20)	Calls	Lat. (s)	Next call	Flagged
S0 no fault	primary model 20	1.0	0.00	3.1-flash-lite	0
S1 transient 429 (once)	primary model 20	2.0	0.00	3.1-flash-lite	0
S2 timeout (once)	primary model 20	2.0	0.00	3.1-flash-lite	0
S3 persistent 429 on primary	fallback model 20	4.0	1.01	3.1-flash-lite	0
S4 model 404 not found	fallback model 20	2.0	0.00	3.6-flash	0
S5 empty response	fallback model 20	2.0	0.00	3.1-flash-lite	0
S6 JSON in markdown fence	primary model 20	1.0	0.00	3.1-flash-lite	0
S7 prose, no JSON	unparsed (raw text) 20	1.0	0.00	3.1-flash-lite	0
S8 all models 500	offline template 20	6.0	2.02	3.1-flash-lite	20

Circuit-breaker probe: 8 consecutive failing calls (fail_max = 5) reached the upstream 5 times; 3 were short-circuited. Sustained 429 on the primary over 8 calls: the primary was tried by 3 calls, then skipped by its model breaker (fail_max = 3, 60 s window).

starts with the configured model and skips a model only while its own circuit breaker is open (three consecutive failures, 60 s). In the repeated fault injection the next call returns to `gemini-3.1-flash-lite` after an empty response and after a persistent 429 (Table IX).

Inert circuit breaker (fixed after the evaluation). The circuit breaker recorded failures but its `allow()` method never raised, so it never short-circuited a call. It now rejects calls while open and lets one trial call through after the window (half-open). The probe now shows 3 of 8 consecutive failing calls short-circuited (0 before), and the per-model breakers mean that a rate-limited primary costs the back-off delay only on the first three calls (Section V-G).

Silent offline fallback (fixed after the evaluation). When every model failed, the provider returned offline template text and the job looked like a normal result. Every step that falls back (offline template, placeholder draft, missing translation) now marks the job as degraded; such a job ends in `needs_review` with an explanatory message. All 20 fault-injection trials in which every model failed are flagged.

Unguarded response shape (fixed after the evaluation). In 7 live runs—every run of one talk (*Elizabeth Gilbert*)—the analysis model returned a JSON array instead of an object, the analysis node raised `'list' object has no attribute 'get'` and the job failed. The failure is deterministic for that input, so retries did not help. Replies are now coerced to an object (an array of objects is unwrapped and merged), and the analysis node re-asks once with an explicit instruction if the expected keys are still missing. With the fix all seven runs of the talk completed, each with a single analysis call, and the talk is included in the ablation and quality results.

Regional endpoints matter. Embedding calls on Vertex's `global` endpoint took about 12 s each from our location versus 0.5–0.7 s on regional endpoints; moving embeddings to `asia-south1` removed RAG indexing from the critical path.

Network resilience of the host. Several runs coincided with machine-wide network outages (DNS failures, SSL errors, stalled connections). Because token refresh in the Google client is synchronous, a stalled refresh blocks the event loop and the per-call timeout cannot fire; one call stalled for over 80 minutes before completing. The fallback STT engine (Gemini API) also failed in these cases because its API key was invalid, and the runner did not fall back a second level. Credentials of every configured engine are now checked at start-up, a rejected engine is skipped, and speech-to-text tries every remaining engine in turn (fixed after the evaluation). Following these stalls we added a 240 s per-chunk timeout to the Vertex STT provider; the end-to-end runs used the version without it, the later cached-stage transcription runs the version with it.

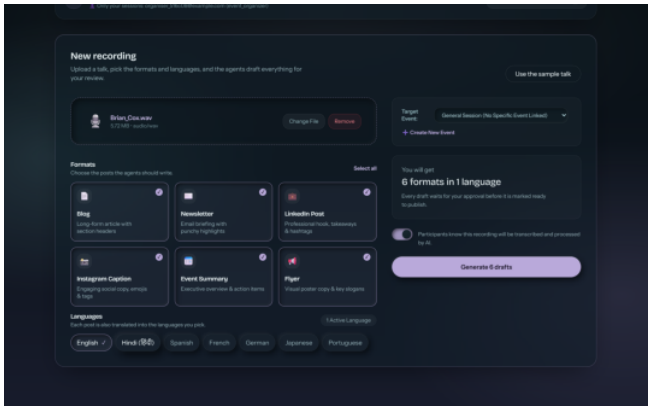
B. Threats to validity

Sample size: 18 talks and 11 long talks are enough for paired non-parametric tests and descriptive statistics, not for fine-grained claims; confidence intervals are wide. *Domain*: TED talks are well-recorded, single-speaker English talks; noisy multi-speaker event audio would raise WER. *Automatic judging*: quality scores come from an LLM judge, not human raters; we use a different, larger judge model, blind conditions, a human reference transcript and a second judge, but judge bias towards fluent text remains. Key points are model-extracted (with verbatim quotes checked mechanically). *Retrieval labels* are produced by construction from model-written questions and may favour paraphrase-friendly queries. *Environment*: latency was measured on a consumer laptop and home network with two concurrent jobs; absolute numbers depend on region and load. *Cost*: we report measured tokens; we could not retrieve official per-token prices for the models used, so monetary cost is not reported.

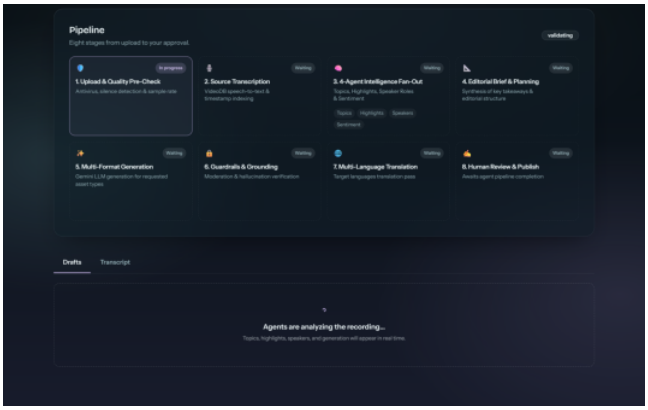
VII. CONCLUSION AND FUTURE WORK

EventAI shows that a recording-to-publication pipeline built from a multimodal LLM, an explicit multi-agent graph, a batched guardrail and selective RAG can turn a 15-minute talk into six review-ready drafts and their translations in about one and a half minutes, at a word error rate of about 2% on real talks. The evaluation supports three practical conclusions. First, parallel writer agents make generation a minor part of latency; transcription and local media analysis dominate. Second, for long talks a writer that sees only the beginning of the transcript under-reports the rest; brief-driven retrieval with modern embeddings fixes this imbalance, but does not by

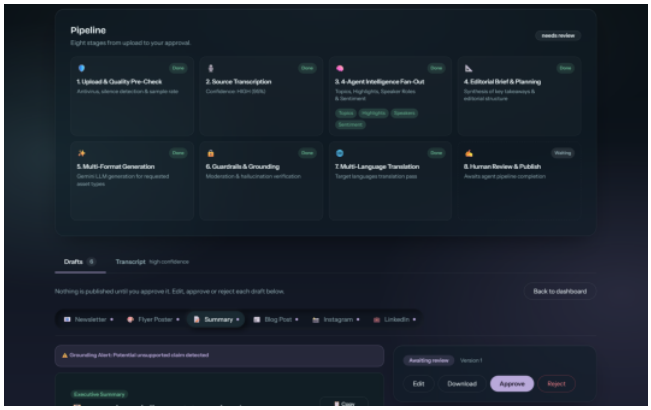
(a) Upload: formats, languages, consent



(b) Live pipeline view (SSE)



(c) Finished job: guardrail alert, review actions



(d) Content Studio: LinkedIn draft

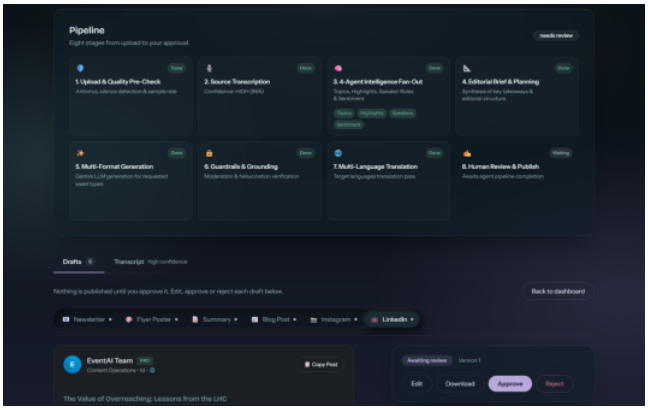


Fig. 12. EventAI web client operated by Playwright on a TED talk: (a) upload with format, language and consent selection; (b) live pipeline view; (c) finished job, where the guardrail flagged one draft and the job awaits review; (d) a LinkedIn draft in the Content Studio.

itself make drafts more complete overall. Third, evaluating the deployed system end to end on real inputs found defects that unit tests with fake providers could not, so live, instrumented evaluation should be part of building such systems.

The defects found this way—sticky model fallback, an inert circuit breaker, silent offline fallback, an unguarded response shape and missing credential checks—have been fixed and the affected experiments re-run. Future work includes: validating every model response against a full per-step schema rather than only its shape; human evaluation of a sample of drafts to calibrate the automatic judge; noisy, multi-speaker and Indian-language recordings; and using event metadata so that flyers can include real logistics instead of "TBA".

Acknowledgment

The evaluation uses the TED-LIUM 3 corpus (CC BY-NC-ND 3.0) for non-commercial academic research. Model calls were made on Google Cloud Vertex AI.

References

[1] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023.

[2] LangChain, Inc., "LangGraph: Build resilient language agents as graphs," https://github.com/langchain-ai/langgraph, 2025, version 0.3.31 used in this work.

[3] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W.-t. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[4] F. Hernandez, V. Nguyen, S. Ghannay, N. Tomashenko, and Y. Estève, "TED-LIUM 3: Twice as much data and corpus repartition for experiments on speaker adaptation," in *Speech and Computer (SPECOM 2018)*, ser. Lecture Notes in Computer Science, vol. 11096. Springer, 2018, pp. 198–208.

[5] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. PMLR, vol. 202, 2023, pp. 28 492–28 518.

[6] S. Gandhi, P. von Platen, and A. M. Rush, "ESB: A benchmark for multi-domain end-to-end speech recognition," *arXiv preprint arXiv:2210.13352*, 2022.

[7] Gemini Team, Google, "Gemini: A family of highly capable multimodal models," *arXiv preprint arXiv:2312.11805*, 2023.

[8] M. Zhong, D. Yin, T. Yu, A. Zaidi, M. Mutuma, R. Jha, A. H. Awadallah, A. Celikyilmaz, Y. Liu, X. Qiu, and D. Radev, "QMSum: A new benchmark for query-based multi-domain meeting summarization," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, 2021, pp. 5905–5921.

[9] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and

P. Liang, "Lost in the middle: How language models use long contexts," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 157–173, 2024.

[10] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 6769–6781.

[11] J. Lee, F. Chen, S. Dua, D. Cer *et al.*, "Gemini embedding: Generalizable embeddings from Gemini," *arXiv preprint arXiv:2503.07891*, 2025.

[12] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[13] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of IR techniques," *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002.

[14] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAS: Automated evaluation of retrieval augmented generation," in *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 2024, pp. 150–158.

[15] S. Yao, J. Zhao, D. Yu, N. Du, I. Shafran, K. Narasimhan, and Y. Cao, "ReAct: Synergizing reasoning and acting in language models," in *International Conference on Learning Representations (ICLR)*, 2023.

[16] Q. Wu, G. Bansal, J. Zhang, Y. Wu, B. Li, E. Zhu, L. Jiang, X. Zhang, S. Zhang, J. Liu, A. H. Awadallah, R. W. White, D. Burger, and C. Wang, "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," *arXiv preprint arXiv:2308.08155*, 2023.

[17] S. Gehman, S. Gururangan, M. Sap, Y. Choi, and N. A. Smith, "RealToxicityPrompts: Evaluating neural toxic degeneration in language models," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 3356–3369.

[18] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, "Judging LLM-as-a-judge with MT-Bench and chatbot arena," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023.

[19] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu, "G-Eval: NLG evaluation using GPT-4 with better human alignment," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023, pp. 2511–2522.

[20] M. T. Nygard, *Release It! Design and Deploy Production-Ready Software*, 2nd ed. Pragmatic Bookshelf, 2018.

[21] F. Wilcoxon, "Individual comparisons by ranking methods," *Biometrics Bulletin*, vol. 1, no. 6, pp. 80–83, 1945.

[22] J. Cohen, "Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit," *Psychological Bulletin*, vol. 70, no. 4, pp. 213–220, 1968.